

A Dirac Operator from the Logarithmic Geometry of Primes

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Abstract

We propose a Dirac-type operator constructed from the logarithmic geometry of prime numbers. Numerical experiments suggest that the spectrum of this operator is strongly correlated with the ordinates of the nontrivial zeros of the Riemann zeta function. This observation motivates a spectral conjecture in the spirit of the Hilbert–Pólya program.

1 Introduction

The nontrivial zeros of the Riemann zeta function

$$\zeta(s)$$

play a fundamental role in number theory.

The famous Riemann Hypothesis states that all nontrivial zeros lie on the critical line

$$\operatorname{Re}(s) = \frac{1}{2}.$$

A long-standing idea, often attributed to Hilbert and Pólya, is that the imaginary parts of the zeros might appear as eigenvalues of a self-adjoint operator.

The aim of this paper is to suggest a simple candidate operator constructed directly from the geometry of prime numbers.

2 Logarithmic Geometry of Primes

Let

$$p_1, p_2, p_3, \dots$$

be the sequence of prime numbers.

We introduce the logarithmic coordinate

$$C = \log p.$$

This transformation has two important properties:

- multiplicative relations become additive,
- growth patterns become approximately linear.

The primes therefore form a discrete set in a logarithmic space.

3 Prime Graph

We define a weighted graph

$$G = (V, E)$$

with

$$V = \{p_1, p_2, \dots, p_N\}.$$

Between two primes we introduce the symmetric weight

$$w(p_i, p_j) = \frac{\log(p_i p_j)}{\sqrt{p_i p_j}}.$$

This coupling combines

- logarithmic geometry,
- scale damping.

4 Dirac-Type Operator

Let

$$Q_{ij} = w(p_i, p_j).$$

We define the operator

$$D = \begin{pmatrix} 0 & Q \\ Q^T & 0 \end{pmatrix}.$$

This matrix is self-adjoint and therefore has a real spectrum

$$D\psi_n = \lambda_n \psi_n.$$

5 Numerical Experiments

Using the first primes and the first Riemann zeros

$$\frac{1}{2} + i\gamma_n$$

we compare the sequences

$$\lambda_n \quad \text{and} \quad \gamma_n.$$

Numerical experiments indicate a strong monotonic correlation.
For moderate matrix sizes the correlation coefficient is approximately

$$\text{corr} \approx 0.97.$$

This suggests a possible spectral relationship.

6 Spectral Conjecture

We propose the following conjecture.

Bamberg Spectral Conjecture

There exists a self-adjoint operator

$$D_{BM}$$

constructed from the logarithmic geometry of primes such that

$$\text{Spec}(D_{BM}) = \{\gamma_n\},$$

where γ_n are the ordinates of the nontrivial zeros of the Riemann zeta function.

7 Interpretation

The proposed construction connects several structures:

structure	interpretation
primes	discrete geometry
operator	dynamics
eigenvalues	spectrum
zeta zeros	resonances

In this picture the zeros of the zeta function appear as spectral quantities of an operator determined by the primes.

8 Outlook

Several questions remain open:

- the precise operator form,
- asymptotic spectral statistics,
- connections to trace formulas.

Further numerical investigations with larger matrices may provide additional insight.

References

- Riemann, B. (1859). On the number of primes less than a given magnitude.
Montgomery, H. The pair correlation of zeros of the zeta function.
Odlyzko, A. Tables of Riemann zeta zeros.